FODS LAB EXPERIMENTS

EXP 1

```
import numpy as np
student_scores = np.array([
    [80, 75, 85, 70],
    [90, 80, 78, 75],
    [70, 85, 80, 80],
    [85, 90, 88, 82]
])
averages = np.mean(student_scores, axis=0)
subjects = ["Math", "Science", "English", "History"]
print("Subject Averages:")
for i in range(4):
    print(subjects[i], "=", averages[i])
highest = np.argmax(averages)
print("Highest Average Subject:", subjects[highest])
print("Highest Average:", averages[highest])
```

```
Subject Averages:
Math = 81.25
Science = 82.5
English = 82.75
History = 76.75
Highest Average Subject: English
Highest Average: 82.75
```

EXP 2

```
import numpy as np
# Product prices
prices = np.array([
    [100, 150, 200],
    [250, 300, 350],
    [400, 450, 500]
])
# Calculate average price
average = np.mean(prices)

print("Average Product Price =", average)
```

```
Average Product Price = 300.0
```

EXP 3

```
import numpy as np

# Columns: Bedrooms, Area, Sale Price
house_data = np.array([
    [3, 1200, 200000],
    [5, 1800, 350000],
    [4, 1500, 280000],
    [6, 2200, 450000],
    [5, 2000, 400000]
])

# Select houses with more than 4 bedrooms
selected_houses = house_data[house_data[:, 0] > 4]

# Calculate average sale price
average_price = np.mean(selected_houses[:, 2])

print("Average Sale Price =", average_price)
```

```
Average Sale Price = 400000.0
```

EXP 4

```
import numpy as np

sales_data = np.array([100000, 120000, 150000, 180000])
total_sales = np.sum(sales_data)

percentage_increase = ((sales_data[3] - sales_data[0])
                       / sales_data[0]) * 100

print("Total Yearly Sales =", total_sales)
print("Percentage Increase =", percentage_increase, "%")
```

```
Total Yearly Sales = 550000
Percentage Increase = 80.0 %
```

EXP 5

```
import numpy as np
fuel_efficiency = np.array([25, 30, 28, 35, 32])
average = np.mean(fuel_efficiency)
model1 = fuel_efficiency[0]
model2 = fuel_efficiency[1]
improvement = ((model2 - model1) / model1) * 100
print("Average Fuel Efficiency =", average, "MPG")
print("Percentage Improvement =", improvement, "%")
```

```
Average Fuel Efficiency = 30.0 MPG
Percentage Improvement = 20.0 %
```

EXP 6

```python
prices = [100, 200, 150]
quantities = [2, 1, 3]
discount_rate = 10
tax_rate = 5
subtotal = 0

for i in range(len(prices)):
    subtotal = subtotal + (prices[i] * quantities[i])
discount = subtotal * discount_rate / 100
price_after_discount = subtotal - discount
tax = price_after_discount * tax_rate / 100
total_cost = price_after_discount + tax

print("Subtotal =", subtotal)
print("Discount =", discount)
print("Tax =", tax)
print("Total Cost =", total_cost)
```

```
Subtotal = 850
Discount = 85.0
Tax = 38.25
Total Cost = 803.25
```